

MVP6 on the Actual-Carrier Endpoint Corridor: A Source-Locked Route Decision after Occurrence and Almost-Prefix Return

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Abstract

TPC-153–161 sharpen both sides of the fixed-shift program. On the structural side, a canonical cut shadow exists, but the frozen artifacts do not determine an actual conservative occurrence lift. This stops only the current-artifacts-only route. The occurrence-augmented route remains unavailable at the first typed object `H1.theorem_backed_occurrence_provenance_crosswalk`; the scalar-plus-eligible-tail route also remains open under its own two-part contract.

On the arithmetic side, a determinant-two two-Möbius periodic core has been advanced from good dyadic blocks to cumulative prefixes outside a sparse dyadic shadow. Exact Abel summation then gives a weighted almost-endpoint interface. This is genuine $L1_{\text{actual}}$ progress. It is not a deterministic all-prefix theorem, a literal physical-weight theorem, or a positive fixed- X -power $L2$ result.

We freeze these facts in MVP6. Invalid data are rejected before seven ordered route verdicts. The audit separately records the typed minimal NOT-TESTABLE blocker antichain and the parent-ready OPEN frontier, preserves complete-route-universe and fresh-registry requirements, and introduces an endpoint-V3 ledger that refuses to charge unknown physical data as zero. The current source-locked decision is

$$\boxed{\text{Verdict}_{153:161} = \text{NOT-TESTABLE}}, \quad \text{FirstMissing} = \frac{\text{H1.theorem_backed_occurrence_}}{\text{provenance_crosswalk}}.$$

No endpoint, prime-pair lower bound, or twin-prime theorem is claimed.

Keywords: fixed shift; occurrence provenance; Möbius correlation; almost endpoint; atomic prefix; route decision.

Claim boundary. An $L1$ theorem on an actual periodic core is not a theorem on the full literal physical carrier. A logarithmic saving has fixed- X -power exponent zero. A sparse exceptional set for continuous logarithmic measure does not control every prescribed atomic endpoint. The scoped current-artifacts stop is not a global architectural impossibility. We do not prove a production occurrence crosswalk, positive $L2$, the strict $1/400$ endpoint, $B_{h_0, \delta}(X) = o(X)$, a prime-pair lower bound, or the twin-prime conjecture.

1 The two corridors entering MVP6

1.1 The structural corridor

TPC-156 identifies the production H1 contract [3]. Its occurrence-augmented map clause requires a theorem-backed provenance crosswalk before the conservative row-separated lift and the literal

downstream maps can be evaluated. Its independent scalar clause requires both a complete original-scale frontier estimate and a theorem-backed disposition of every eligible-tail-open path. In symbols,

$$\begin{aligned}\text{Map}_{H1} &= \text{Crosswalk} \wedge \bigwedge_{\star \in \mathcal{D}} (D_\star = 0) \wedge \text{RegistryComplete}, \\ \text{Scalar}_{H1} &= \text{FUM}_{o(X)} \wedge \text{ETO}_{\text{disposition}}, \\ \text{H1Complete} &= \text{Map}_{H1} \vee \text{Scalar}_{H1},\end{aligned}$$

where

$$\mathcal{D} = \{L, Q_D, Q_Z, G, P, DZ, GP, \text{cover}, \text{rec}\}.$$

The occurrence registry is an additional totality requirement, not a tenth algebraic defect silently merged into this vector.

The finite archive has a canonical conservative cut shadow. It also has a rigorous nonuniqueness result for completions using only current fields. Consequently,

$$\text{current artifacts only} \quad \longmapsto \quad \text{STOP-ROUTE}$$

in that exact cell. It does not imply that an enriched theorem-backed occurrence carrier cannot exist. Therefore neither the occurrence-augmented map route nor the scalar-plus-ETO route is globally stopped.

1.2 The arithmetic corridor

TPC-159 imports the quantitative affine-correlation input of Tao–Teräväinen and its Möbius nonpretentiousness ancestry [1, 2, 4]. For the determinant-two periodic core, take $a, s, R \in \mathbb{N}$, $(a, s) = 1$, as odd, and $d, u \in \mathbb{Z}$ satisfying $su - ad = 2$. For $z \in \mathbb{Z}$, write

$$q = as, \quad t(z) = ad + qz, \quad c_z = \mu(d + sz)\mu(u + az),$$

and let ρ be a bounded R -periodic function. TPC-159 proves that outside a dyadic shadow $\mathcal{S}_{X,J}$,

$$\frac{q}{T} \left| \sum_{0 < t(z) \leq T} c_z \rho(z) \right| \ll \|\rho\|_\infty \left\{ (\log X)^{-\kappa_0} + 2^{-J} + \frac{q}{T} \right\}, \quad (1)$$

provided $2^J \sqrt{X} \leq T \leq X$, while

$$\frac{1}{\log X} \int_{\mathcal{S}_{X,J}} \frac{dT}{T} \ll J(\log X)^{-\kappa_0}. \quad (2)$$

The geometric dyadic decomposition incurs no multiplicative J loss in [equation \(1\)](#). The level is

$$\text{PROVED_L1_ACTUAL_PREFIX_ALMOST_ENDPOINT.}$$

TPC-160 then applies exact Abel summation [5]. If $A_i = \sum_{j \leq i} \sigma_j$ and $d_i = w_i - w_{i+1}$, with the terminal convention $d_m = w_m$, then

$$\sum_{i=1}^m \sigma_i w_i = \sum_{i=1}^m A_i d_i. \quad (3)$$

Splitting the variation into good and exceptional endpoint cells gives

$$\frac{q}{T} \left| \sum_i \sigma_i w_i \right| \ll \|\rho\|_\infty \{ \varepsilon_X V_{\text{good}}(w) + (1 + q/T) V_{\text{bad}}(w) \}, \quad (4)$$

where $\varepsilon_X = (\log X)^{-\kappa_0} + 2^{-J} + q/T$. This is an exact weighted interface, but the literal physical weight, actual endpoint registry and production bad-variation estimate remain unavailable.

For a prefix cutoff, the Abel derivative is one unit atom at the chosen endpoint. A set of zero logarithmic measure may contain that atom. Thus [equation \(2\)](#) alone cannot promote [equation \(1\)](#) to every deterministic prefix.

2 A source-locked MVP6 snapshot

TPC-161 supplies the typed integration manifest used below [6].

Definition 2.1 (Valid snapshot). An MVP6 snapshot is valid if:

- (a) the TPC-161 manifest, schema, audit and generating script match their canonical UTF-8/LF hashes;
- (b) direct TPC-156 and TPC-160 consistency anchors match the structural and arithmetic exports used by the classifier;
- (c) all hashes have integrity semantics only;
- (d) node identifiers, dependency modes, roles, scopes, carriers, normalizations and registries satisfy their schemas;
- (e) the active typed DAG is acyclic;
- (f) every stop, route-universe-completeness claim and reroute crosswalk resolves through the route-evidence registry to a typed source record with matching scope and route endpoints;
- (g) the route table exactly covers its declared universe, and every stopped cell carries source, scope, carrier, normalization, registry and complete-cell coverage;
- (h) every endpoint-V3 rational value has positive denominator and every pass/stop direction is consistent; and
- (i) all claim-boundary Booleans obey the achieved levels.

Any failure returns the outer result INVALID. Source drift makes the certificate stale; it does not make a mathematical theorem false.

3 Two frontiers that must not be merged

Let $\mathcal{A}$ be the active ancestor closure of the selected root. Define

$$\mathcal{B}_{\text{NT}} = \min_{\preceq} \{v \in \mathcal{A} : \text{status}(v) = \text{NOT-TESTABLE} \text{ or } v \text{ is scope-incompatible}\}. \quad (5)$$

This is a typed blocker antichain: descendants are suppressed by their earliest missing ancestors. The audit computes each recorded transitive ancestor set directly from the source DAG rather than trusting a submitted parent label. A proved node with incompatible scope is therefore also a blocker, exactly as in [equation \(5\)](#).

Separately define the parent-ready open frontier

$$\mathcal{F}_{\text{open}} = \{v \in \mathcal{A} : \text{status}(v) = \text{OPEN}, v \text{ is scope-compatible, every active parent of } v \text{ is proved}\}. \quad (6)$$

A node in $\mathcal{F}_{\text{open}}$ is a live mathematical target. A node in $\mathcal{B}_{\text{NT}}$ needs a missing typed artifact before its predicate is testable. The two sets are disjoint by their combined status-and-scope typing, and neither may be used as a substitute for the other.

Proposition 3.1 (Current typed frontiers). *For the selected synthesis root,*

$$\mathcal{B}_{\text{NT}} = \left\{ \begin{array}{l} \text{H1.theorem_backed_occurrence_provenance_crosswalk,} \\ \text{H9.endpoint_registry, H9.literal_weight_registry,} \\ \text{H9.normalization_registry, H9.phase_cell_registry} \end{array} \right\}. \quad (7)$$

The canonical selected first missing representative is the H1 crosswalk. The separate parent-ready frontier is

$$\mathcal{F}_{\text{open}} = \left\{ \begin{array}{l} \text{O161.bad_endpoint_pointwise_core,} \\ \text{O161.direct_additive_twist_core} \end{array} \right\}. \quad (8)$$

Proof. The canonical cut occurrence shadow, witness-verifier contract and typed H1 decision are proved. Their next node requires a theorem-backed production crosswalk and is NOT-TESTABLE. Independently, the literal-weight, phase-cell, endpoint and normalization registries are missing roots of the physical-return branch. None of these five nodes descends from another, while the lift, literal downstream maps, physical cover, reconnection and complete physical registry descend from them. This proves [equation \(7\)](#); canonical node ordering selects the H1 entry first. The two nodes in [equation \(8\)](#) are status-OPEN, have ready artifacts and proved parents, so they remain a distinct live frontier. $\square$

4 Route semantics and classifier

The source-locked route universe contains five declared cells:

R_{archive}	: actual lift derivable from current fields only,
$R_{\text{map+}}$	: occurrence-augmented theorem-backed map route,
$R_{\text{scalar+ETO}}$	: complete FUM scalar theorem plus ETO disposition,
R_{twist}	: direct additive-twist core route,
$R_{\text{bad endpoint}}$	: pointwise control on exceptional prefix endpoints.

Only R_{archive} is stopped. Since no independent theorem proves these cells form a complete universe of possible arguments, this cannot imply ARCHITECTURE-INFEASIBLE.

A reroute is legal only when the selected route has an exact stop, the alternative is open, a theorem-backed typed crosswalk connects their objects and the alternative has a genuinely fresh registry. Renaming a registry or reusing the stopped route's rows makes the snapshot INVALID, not REROUTE.

The production classifier runs in SOURCE_LOCKED mode. Each route stop is resolved to its scoped source result, while any future completeness theorem or crosswalk must be resolved by canonical repository path and hash. The reachability suite instead runs in the disjoint SYNTHETIC_REACHABILITY mode. Its two file-backed predicate fixtures and inline stopped-cell assumptions explicitly carry NONE_SYNTHETIC_REACHABILITY_ONLY semantics; they are not theorem evidence and cannot be imported into the production mode.

Definition 4.1 (Ordered valid-snapshot verdicts). After validation, the first satisfied predicate in the following list is returned:

$$\begin{aligned} \text{GO, ARCHITECTURE-INFEASIBLE, REROUTE, STOP-ROUTE,} \\ \text{NOT-TESTABLE, ARITHMETIC-FRONTIER, OPEN.} \end{aligned} \tag{9}$$

- (i) GO: every active requirement is proved in matching scope and the full endpoint has strict positive slack.
- (ii) ARCHITECTURE-INFEASIBLE: an independent theorem proves the route universe complete and every route has an exact complete-cell stop.
- (iii) REROUTE: the selected route is stopped and a valid fresh typed alternative is selected.
- (iv) STOP-ROUTE: the selected route is stopped and no valid alternative is selected.
- (v) NOT-TESTABLE: the typed blocker antichain is nonempty.
- (vi) ARITHMETIC-FRONTIER: every structural node and the physical endpoint are proved, and all remaining unresolved nodes are scope-matched open positive-L2 arithmetic targets.
- (vii) OPEN: every other valid state.

Theorem 4.2 (Classifier totality). *Validation followed by [equation \(9\)](#) returns exactly one of*

$$\begin{aligned} \text{INVALID, GO, ARCHITECTURE-INFEASIBLE, REROUTE,} \\ \text{STOP-ROUTE, NOT-TESTABLE, ARITHMETIC-FRONTIER, OPEN.} \end{aligned}$$

Every result is reachable on a finite typed synthetic assumed-predicate regression fixture.

Proof. Validation either fails, giving the separate outer result, or produces a finite valid state. The ordered list has a final catch-all, hence the first true predicate is unique. The regression suite constructs one logical witness for each verdict and one invalid route/registry witness for INVALID. This establishes classifier totality only; it asserts none of the synthetic predicates as mathematics. $\square$

5 Endpoint V3

Endpoint V3 asks six literal questions before arithmetic and physical budgets may be combined:

V3.1 Are the coefficients the literal physical coefficients?

V3.2 Is the shift the fixed physical $h_0 = 2$?

V3.3 Is the atomic normalization the physical one?

V3.4 Is the representation canonical or proved minimal?

V3.5 Is the actual active support completely registered?

V3.6 Does the physical loss satisfy a strict upper bound below $1/400$?

An unknown answer is not zero cost.

The arithmetic ledger records the strongest achieved theorem and the target separately:

$$\sigma_{\text{achieved}}^{(X)} = 0, \quad \sigma_{\text{target}}^{(X)} \geq \frac{1}{400}. \quad (10)$$

The zero in [equation \(10\)](#) means that the proved saving is logarithmic, not that there is no arithmetic progress. The physical ledger requires

$$\Lambda_{\text{phys}} < \frac{1}{400}, \quad (11)$$

with a complete independent registry. Full synthesis additionally requires compatible scopes and strict net slack. Current artifacts certify neither [equation \(10\)](#)'s fixed-power target nor [equation \(11\)](#).

Proposition 5.1 (No pseudo-endpoint). *None of the following alone is an endpoint-V3 pass: an almost-endpoint log saving, continuous exceptional-set density, a conditional literal-weight interface, a finite empty ETO census, or an upper cost bound at or above $1/400$.*

Proof. They respectively fail fixed- X -power, atomic endpoint, production weight, growing-scale coverage and strict inequality requirements. $\square$

6 The MVP6 decision

Theorem 6.1 (Source-locked MVP6 verdict). *For the TPC-153–161 source-locked state,*

$\text{Verdict}_{153:161} = \text{NOT-TESTABLE}, \quad \text{FirstMissing} = \frac{\text{H1.theorem_backed_occurrence_}}{\text{provenance_crosswalk}}.$

The current-artifacts-only route is stopped in its exact declared cell; the selected augmented map route and scalar-plus-ETO route are not globally stopped.

Proof. All source, type, DAG and endpoint validations pass, so the snapshot is not INVALID. The selected augmented map route has no stop. The declared route universe has no completeness theorem, excluding ARCHITECTURE-INFEASIBLE, and no alternative is activated with a fresh registry, excluding REROUTE. By [proposition 3.1](#), the selected active set has a five-node minimal NOT-TESTABLE antichain whose canonical first entry is the H1 crosswalk. The precedence therefore returns NOT-TESTABLE.

Independently, TPC-159 supplies an actual periodic-core almost-endpoint L1 theorem and TPC-160 a weighted interface, but the literal physical weight, phase-cell, normalization and endpoint registries, every atomic prefix and fixed positive X -power remain unavailable. Thus ARITHMETIC-FRONTIER and GO would also be false even if they were tested before the catch-all. $\square$

Corollary 6.2 (Precise progress label). *The current state contains structural L1, actual periodic-core almost-endpoint L1, and an exact weighted-return interface. The achieved fixed- X -power level remains below $L2_{\text{actual-positive}}$.*

7 Next forced objects

Structural object. The selected route must next provide

H1.theorem_backed_occurrence_provenance_crosswalk.

It must cover every nonsoft ETO and FUM path, separate each target occurrence, conserve source columns and preserve theorem source, canonical parent, stage, multiplier, native value, fixed shift, physical normalization and support lineage. Only then can the actual lift and nine defects be evaluated.

Arithmetic/physical-return objects. The parallel corridor must provide a source-locked literal physical weight, phase-cell registry, actual endpoint registry and literal normalization registry. It must then prove either small bad variation in [equation \(4\)](#) or a pointwise theorem on the exceptional atomic endpoints. These are the next objects needed to turn TPC-159/160 into a literal physical-return statement. A further theorem would still be required to replace the logarithmic saving by a positive fixed power of X .

Decision discipline. If the crosswalk is constructed, the first missing pointer moves forward to the first nonzero or unevaluable literal map defect. If a theorem proves that no crosswalk exists on the actual carrier, the augmented map cell may receive a typed stop. If the scalar-plus-ETO clause is completed, it becomes a legitimate reroute only with its own fresh registry and crosswalk. None of these future outcomes is prejudged here.

8 Reproducible audit

The standard-library generator dynamically reads the TPC-161 manifest rather than embedding its hash. It records canonical hashes for the TPC-161 manifest, schema, audit and script, and directly checks the TPC-156 scoped-stop and TPC-160 weighted-interface anchors. It also locks the two synthetic predicate fixtures used only by the reachability suite, then emits a strict snapshot and audit.

Mutation regressions reject:

1. a malformed snapshot before mathematical classification;
2. an unregistered or fabricated theorem label, or classifier source-hash drift;
3. route-universe completeness without a registered independent theorem;
4. architecture infeasibility from stops on only the declared sample of routes;
5. rerouting without a fresh registry or registered typed crosswalk;
6. promotion of the current-artifacts stop to actual-carrier impossibility;
7. collapsing the five-node selected-root blocker antichain to its first reporting representative, or falsifying its transitive ancestry;
8. dropping a scope-incompatible blocker or merging an NOT-TESTABLE blocker with a parent-ready OPEN node;
9. assigning TPC-159 a deterministic all-prefix conclusion;
10. promoting logarithmic decay to a positive X -power;

11. treating exceptional logarithmic density as atomic avoidance;
12. promoting the TPC-160 interface to a production physical-weight theorem; and
13. an endpoint-V3 pass with any unknown literal field or without a strict $1/400$ inequality.

9 Conclusion

MVP6 finds that both corridors remain live but not yet joined. The selected structural first object is now one theorem-backed provenance crosswalk, not an unspecified missing matrix. Four independent physical-return registries form the rest of the selected-root minimal blocker antichain. The arithmetic theorem now reaches almost every cumulative endpoint of the actual periodic core, not merely one dyadic block. The remaining gap is explicit: construct the actual carrier and return the arithmetic estimate through literal weights, phases, normalizations and atomic endpoints. Until that happens, NOT-TESTABLE is the only source-compatible verdict.

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